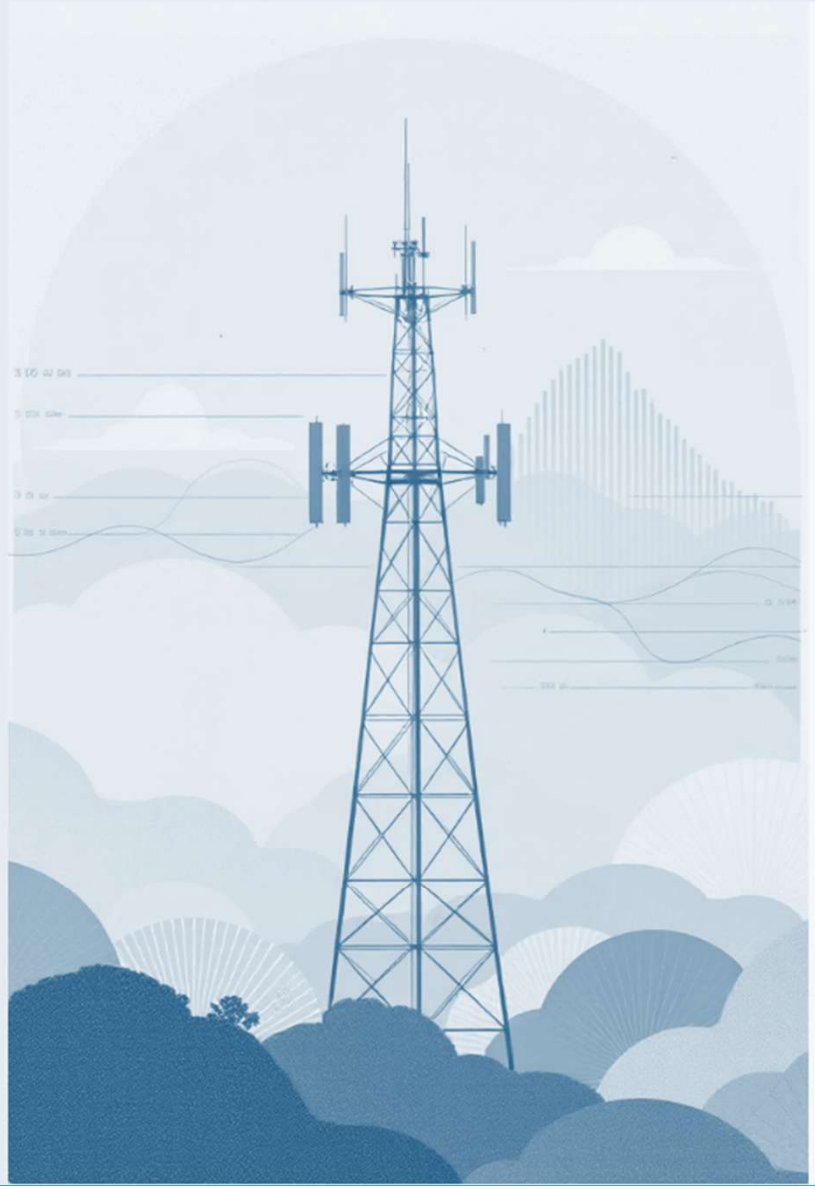


Telco Customer Churn EDA Project

Team: Group 3 | **Date:** 15th March 2026

DATASET INTRODUCTION

- **Dataset chosen:**
  - **Source:** Telco Customer Churn CSV (WA_Fn-UseC_-Telco-Customer-Churn.csv)
  - **Size:** 7,043 customer records across 21 variables
  - **Target Variable:** Churn (Yes/No)
  - **Key Features:**
    - **Demographics:** gender, SeniorCitizen, Partner, Dependents
    - **Account Information:** customerID, tenure (months with company)
    - **Services:** PhoneService, MultipleLines, InternetService (DSL/Fiber optic/No), OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
    - **Billing:** Contract (Month-to-month/One year/Two year), PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges

DATA IMPORT AND INITIAL INSPECTION

Data Import and Initial Inspection

The dataset was successfully loaded using pd.read_csv(), creating a pandas DataFrame with 7,043 records and 21 variables. Initial inspection confirmed the presence of mixed data types including numeric features (tenure, MonthlyCharges) and categorical variables (Contract, PaymentMethod).

Key Inspection Methods

•df.shape - Verified dimensions

•df.head() - Previewed structure

•df.info() - Checked data types

•df.describe() - Summary statistics

DATA CLEANING

1. Missing Value Detection - Identified 11 rows with missing TotalCharges requiring handling

2. Data Type Conversion - Converted TotalCharges to numeric using pd.to_numeric() with error coercion

3. Record Removal - Dropped incomplete entries with dropna(), resulting in 7,032 valid records

4. Duplicate Check - Validated no duplicate records exist using df.duplicated().sum()

Feature Engineering

Created ServiceCount by identifying 9 service-related features and calculating subscription totals. This derived metric enables analysis of service bundling patterns and cross-selling opportunities.

EXPLORATORY DATA ANALYSIS (EDA)

Descriptive Statistics

32	$65	$2,283	3
Average Tenure	**Monthly Charges**	**Total Revenue**	**Services Used**
Months with company	Average subscription cost	Per customer lifetime value	Average number of services

These baseline statistics establish customer behavior patterns before deeper churn analysis.

Feature Distribution Analysis

Tenure Distribution

Mean: 32.42 months | Median: 29 months | Maximum: 72 months

Many customers leave within first few years, indicating higher churn risk among new customers.

Monthly Charges

Mean: $44.79 | Median: $70.35 | Range: $18.25-$118.75

Wide pricing variation suggests different service packages influence churn behavior.

Service Count

Mean: 3.36 services | Maximum: 8 services

Most customers subscribe to multiple services, indicating cross-selling opportunities.

Short Tenure = High Churn

New customers in first 24 months show highest attrition risk

Month-to-Month Dominates

Flexible contracts increase churn probability vs long-term

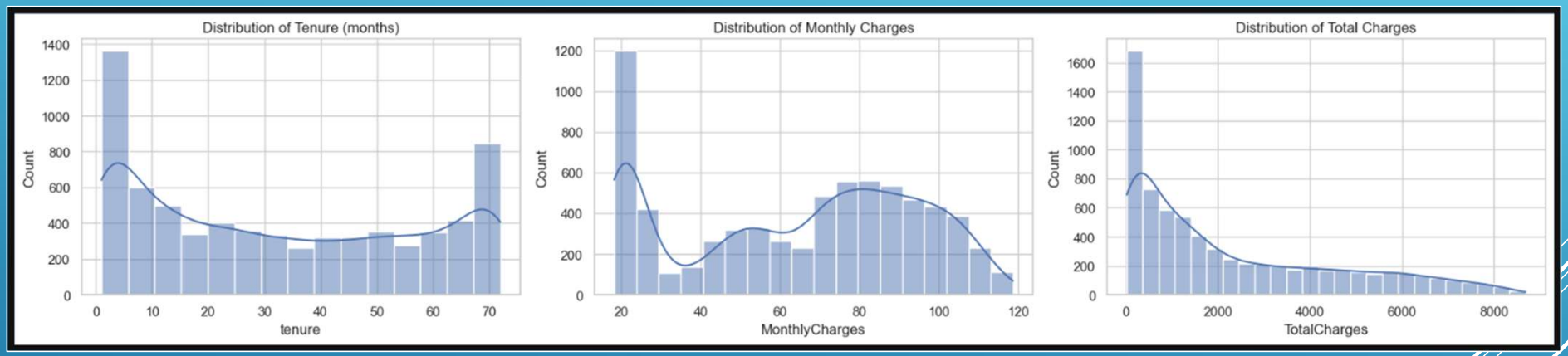
Higher Charges Increase Churn

Churners pay more monthly, suggesting price sensitivity

Service Bundling

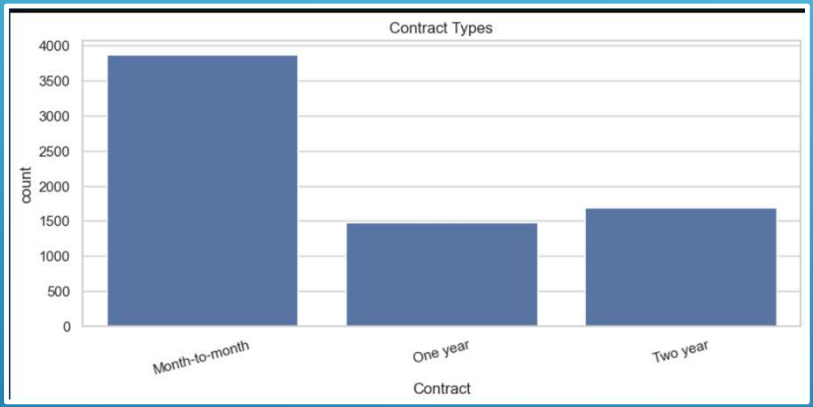
Most customers subscribe to multiple services (average 3.36)

EXPLORATORY DATA ANALYSIS (EDA)
DATA VISUALISATION



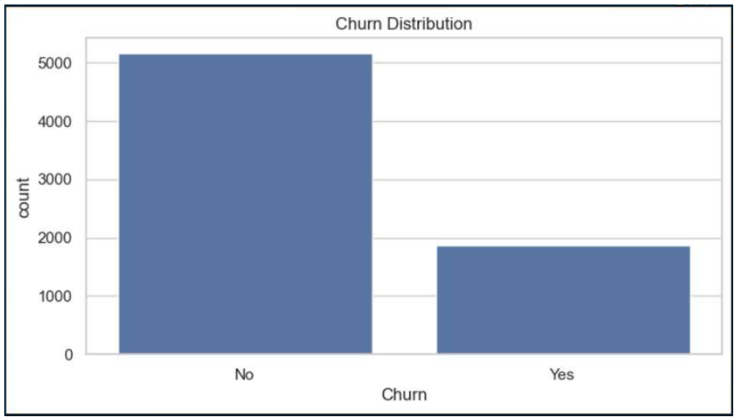
- A large proportion of customers leave early in their lifecycle, indicating that new customers are more likely to churn.
- The wide variation in pricing suggests customers subscribe to different bundles of telecom services, which may influence churn behavior.

EXPLORATORY DATA ANALYSIS (EDA)
DATA VISUALISATION

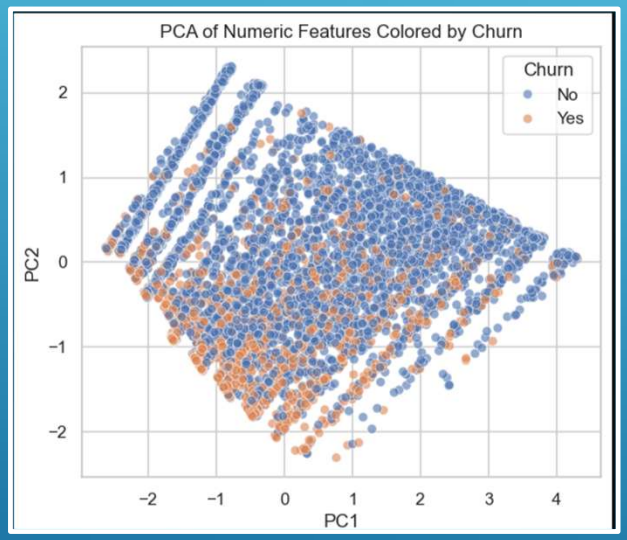


Short-term contracts provide flexibility for customers, but they also increase the likelihood of churn compared to long-term contracts.

Higher service costs may lead to customer dissatisfaction or price sensitivity, increasing the probability of churn.



EXPLORATORY DATA ANALYSIS (EDA)
DATA VISUALISATION



Customers who stay longer are less likely to churn, while higher monthly charges slightly increase churn probability, making these key variables for churn prediction.

This visualization helps identify whether churned and non-churned customers form distinct clusters, indicating that the underlying numeric features contain patterns useful for predicting customer churn.

EXPLORATORY DATA ANALYSIS (EDA)
GROUP ANALYSIS

Group Analysis Findings

Contract Type	Churn Rate
Month-to-month	42.7%
One year	11.2%
Two year	2.8%

Group-by operations revealed clear differences in churn behavior across customer segments. Fiber optic customers on month-to-month contracts have highest churn rate. Contract type and service type strongly influence churn behavior.

EXPLORATORY DATA ANALYSIS (EDA)
FEATURE ANALYSIS

Analysis Findings

Methodology

Correlation analysis on 7,032 customers across tenure, charges, contract type, and 9 service categories.

Key Results

Tenure correlation: -0.35. Monthly charges: +0.19. Fiber Optic/Month-to-Month: 42.7% churn vs 2.8% for Two-Year contracts. New customers (0-12 months) account for 54% of all churn.

Feature Correlations

Tenure <6 months churns at 47% vs 6% for 24+ months. Month-to-month contracts at >$80/month reach 58% churn. Service bundles reduce churn: 4+ services (15%) vs single service (45%).

Actionable Insights

Prioritize the first 6 months. Month-to-month is 15x riskier than long-term. Service bundling is a key retention lever, reducing churn ~8% per additional service.

ADVANCED PYTHON TECHNIQUES

Lambda Functions

What was done
Used lambda functions to perform simple transformations, converting the categorical Churn variable into a numeric flag and grouping tenure into lifecycle stages.

Key Result
Created new features:
ChurnFlag (0 = No churn, 1 = Churn) for statistical analysis
TenureGroup (New, Recent, Established, Loyal) for customer segmentation

Insight
Lambda-based transformations simplified feature engineering and revealed that new customers have a higher churn risk compared to loyal customers.

User-Defined Functions

What was done
Custom Python functions were created to automate repetitive tasks such as generating dataset summaries, counting services, and detecting outliers using the IQR method.

Key Result
Reusable functions enabled automated data profiling, outlier detection, and efficient analysis across multiple columns.

Insight
User-defined functions improved code efficiency, readability, and reusability, making the EDA workflow more scalable and consistent.

List Comprehensions

What was done
Used list comprehension to efficiently identify specific columns, such as service-related features and columns with missing values.

Key Result
Automatically extracted relevant columns without manually specifying each one.

Insight
Improved code efficiency, readability, and scalability for processing datasets with many variables.

INSIGHTS AND CONCLUSIONS

High-Risk Segment

Customers with short tenure and month-to-month contracts represent highest churn risk.

Service Impact

Fiber optic internet users exhibit higher churn rates compared to DSL or no internet service.

Price Sensitivity

Customers who churn tend to have higher monthly charges, suggesting price sensitivity.

Next Steps

- Build predictive churn models (Logistic Regression, Random Forest)
- Perform customer segmentation and lifetime value analysis
- Investigate service quality and customer support interactions
- Conduct cohort or time-based analysis of churn trends